Computer Networks Cheat Sheet

Chapter 1 - Top Down Approach

1. Basic Terminologies

Term	Definition
Network	Group of connected devices sharing data.
Latency	Time for data to travel from source to destination.
Bandwidth	Maximum data transfer rate (bits per second).
Throughput	Actual data transfer rate (usually less than bandwidth).
Node/Host	Any device (PC, phone, server) connected to a network.
Network Edge	End systems (PCs, phones) connected to the internet.
Network Core	Interconnected routers and switches (network of networks).
QoS	Quality of Service: Prioritizes critical traffic (e.g., video calls).

2. Types of Networks

Type	Definition	Example
LAN	Local Area Network: Connects devices in a small area (home, office).	Wi-Fi, Ethernet
WAN	Wide Area Network: Connects LANs over large distances.	Internet
Access Network	Connects end systems to the internet (DSL, Cable, Wi-Fi, Cellular).	Home Wi-Fi, 4G/5G

3. Access Networks

A. Residential Access

- **DSL:** Uses `copper phone lines`. Asymmetric: faster download than upload.
- **Cable Internet:** Uses `coaxial cables`. Shared bandwidth: speed drops during peak times.

- **FTTH:** Fiber To The Home: Uses `fiber optic cables`. Highest speed residential option.

B. Institutional/Enterprise Access

- **Ethernet:** Wired LAN technology (high speed, reliable).
- **Wireless LAN (Wi-Fi):** Uses `802.11 standard` (Wi-Fi 5/6).

C. Mobile Access

- **3G/4G/5G:** Uses cellular towers. `5G` offers faster speeds and lower latency.
- **LTE:** Long Term Evolution: Standard for high-speed wireless communication.

4. Physical Media

Guided Media (Wired)

Type	Description	Example
Twisted Pair	Two copper wires twisted to reduce interference.	Ethernet cables
Coaxial Cable	Single copper wire with shielding. Used for cable TV/internet.	Cable internet
Fiber Optic	Uses light pulses for high-speed, long-distance data transfer.	Internet backbone

Unguided Media (Wireless)

Type	Description	Example
Radio Waves	Wireless signals (Wi-Fi, Bluetooth, 4G/5G).	Wi-Fi, 5G
Satellite	Communication via satellites in orbit.	GPS, Satellite internet
Microwave	Point-to-point terrestrial communication.	Long-distance telephone

5. Network Core Concepts

Switching Methods

Method	Description	Use Case
Circuit Switching	Dedicated path for the entire session.	Traditional phone calls
Packet Switching	Data split into packets; shared network resources.	Internet, Modern networks

Message Switching	Store-and-forward entire messages.	Older systems, Email

Network Core Functions

Function	Description
Forwarding	Router moves packet from input to output port (local action).
Routing	Determines the best path for packets (global process).
Packet Scheduling	Determines order of packet transmission.

6. Delays in Networks

Types of Delay

Delay Type	Description	Formula
Processing	Time to check packet header and determine route.	Typically negligible
Queuing	Time waiting in router's buffer.	Variable (depends on traffic)
Transmission	Time to push packet onto the link.	`L/R` (L=packet size, R=link rate)
Propagation	Time for a bit to travel from sender to receiver.	`d/s` (d=distance, s=speed)

Total Delay = Processing + Queuing + Transmission + Propagation

Delay Calculations

Example: A 1500-byte packet over a 1 Gbps link, 1000 km distance, propagation speed 2×10^8 m/s.
Transmission Delay = $(1500\times8$ bits$) / 10^9$ bps = 12 μs
Propagation Delay = 10^6 m / $(2\times10^8$ m/s$)$ = 5 ms
Total Delay ≈ 5.012 ms (ignoring processing and queuing)

Key Concept: End-to-End Delay
For a path with N links and N-1 routers:
End-to-End Delay = N×(Transmission Delay) + Total Propagation Delay + Total Queuing Delay

7. Throughput & Bottleneck

Throughput Concepts

- **Throughput:** Actual data transfer rate between source and destination.
- **Instantaneous Throughput:** Rate at a specific moment in time.
- **Average Throughput:** Total data transferred divided by total time.
- **Bottleneck Link:** The slowest link in the end-to-end path.

$$\text{Throughput} = \min(R_1, R_2, \ldots, R_n)$$

Example: If links have rates 10 Mbps, 100 Mbps, and 1 Gbps, the throughput is limited to 10 Mbps.

Bandwidth-Delay Product

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Bandwidth-Delay Product = Bandwidth × Round-Trip Time
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Represents the maximum amount of data that can be "in flight" in the network.

Network Performance Metrics

- **Goodput:** Application-level throughput (excluding headers)
- **Jitter:** Variation in packet delay
- **Packet Loss:** Percentage of packets that fail to arrive

8. Protocol Stack & Layering

TCP/IP Model (5 Layers)

Layer	Protocols	Function
Application	HTTP, FTP, SMTP, DNS	User applications
Transport	TCP (reliable), UDP (fast)	End-to-end communication
Network	IP, Routing protocols	Addressing & routing
Link	Ethernet, Wi-Fi	Local network communication
Physical	Cables, radio waves	Bit transmission

OSI Model (7 Layers)

Layer	Function
7. Application	User interface (HTTP, FTP)

6. Presentation	Data format (encryption, compression)
5. Session	Manages connections
4. Transport	End-to-end communication (TCP/UDP)
3. Network	Routing (IP)
2. Data Link	Framing (Ethernet, Wi-Fi)
1. Physical	Bit transmission (cables, radio)

9. Encapsulation

Data Encapsulation Process

Data is wrapped with protocol information at each layer:

1. **Application:** Original data (message)
2. **Transport:** Segment (TCP/UDP header + data)
3. **Network:** Datagram/Packet (IP header + segment)
4. **Link:** Frame (MAC header + packet + trailer)
5. **Physical:** Bits (converted to signals)

Protocol Data Units (PDUs)

- Transport Layer: Segment
- Network Layer: Packet/Datagram
- Link Layer: Frame
- Physical Layer: Bits

Network Devices & Layers

Device	OSI Layer	Function
Hub	Physical (Layer 1)	Connects devices, broadcasts data
Switch	Data Link (Layer 2)	Forwards frames based on MAC addresses
Router	Network (Layer 3)	Routes packets between networks
Modem	Physical (Layer 1)	Modulates/demodulates signals

10. Network Security

Security Threats

Threat	Description	Solution

Packet Sniffing	Capturing data packets (e.g., passwords).	Encryption (HTTPS, VPN)
IP Spoofing	Faking source IP address.	Ingress filtering, Authentication
DDoS	Overloading a server with traffic.	Firewalls, rate limiting, CDNs
Man-in-the-Middle	Intercepting communication between parties.	Encryption, Authentication

Security Mechanisms

- **Encryption:** Scrambling data to prevent unauthorized access
- **Authentication:** Verifying the identity of users/devices
- **Firewalls:** Filtering network traffic based on security rules
- **VPN:** Virtual Private Network for secure remote access
- **IDS/IPS:** Intrusion Detection/Prevention Systems

Defense in Depth
Using multiple layers of security controls to protect networks and data.

11. Internet Structure

Internet Hierarchy

- **Tier 1 ISPs:** Global reach, don't pay for transit (e.g., AT&T, Verizon)
- **Tier 2 ISPs:** Regional providers, connect to Tier 1 ISPs
- **Tier 3 ISPs:** Local access providers (e.g., local cable companies)
- **IXPs:** Internet Exchange Points where ISPs peer with each other
- **Content Providers:** Companies like Google, Netflix with their own networks

Internet Standards

- **IETF:** Internet Engineering Task Force
- **RFC:** Request for Comments (standards documents)
- **IEEE:** Institute of Electrical and Electronics Engineers
- **ISO:** International Organization for Standardization

Protocol Development
Standards are developed through consensus and implemented in products and services.